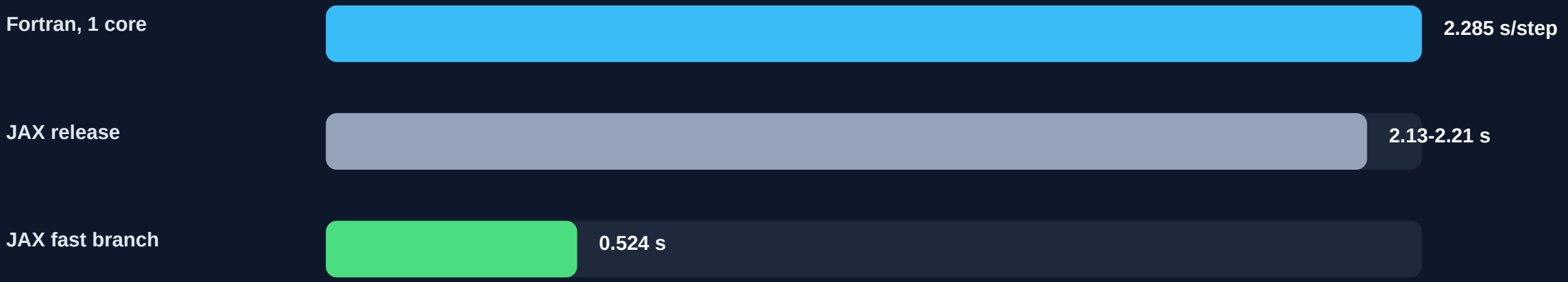
ROCKE-3D / JAX: benchmark state at a glance

Steady-state speed is now compelling on the fast candidate; qualification, startup and fidelity remain the constraints.



Bars show steady-state mean; release range is two V100 nodes; fast branch is one matched run.

4.4x

Fortran / fast JAX

2.285 / 0.524; steps 2-107.

933 + 96 s

Fast JAX startup

Step 0 then step 1; compile-heavy.

~580 steps

Estimated payback

About 12 days at 48 steps/day; arithmetic, not measured.

Claim boundary: 4.4x applies only to the optimized fast-branch steady steps, not to short-run wall time, the release configuration, other dates or a validated long free run.

Sources: RELEASE_NATIVE_RADIATION_20261012; SPEED_STEP_TIME_LOG_20261012 s13-14

How the current coupled step is assembled

The production path is a mixed host/device program; fast-path flags change selected stages without changing defaults.

STATE + SETUP

Restart / carried state
Host providers

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ATMOSPHERE

JAX dynamics
CONDSE / convection

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SURFACE + LAND

JAX tiles, PBL, GHY
Ent and daily updates

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OCEAN + ICE

JAX ocean / sea ice
host pole / callbacks

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RADIATION

native JAX SOCRATES
inside coupled step

RELEASE CONFIGURATION

+ All behavior flags off by default.
+ Native JAX radiation integrated; standalone/input-removal flags are opt-in.
+ Measured steady-state parity on two V100 nodes (2.13-2.21 vs 2.24-2.28 s/step).

FAST BRANCH CONFIGURATION

+ Multiple opt-in flags; cloud-base/MSTCNV runs in persistent Numba worker; ENT_BATCH, pole, dynamics, host I/O tuned.
+ Exact matched run: Fortran x2 then JAX on the same exclusive V100; 108-step window.
+ Worker uses a separate Numba-capable Python environment; deployment packaging is unresolved.

Input boundary: same nov26 initial state plus five step-0 Fortran dumps (~278 MB); no per-step records in the five-day standalone-capable run. This is not a restart-only cold launch.

Sources: JAX_COVERAGE_MATRIX; RELEASE_NATIVE_RADIATION_20261012; STANDALONE_INPUT_AUDIT s9-10; SPEED_RUN_RECIPE_20261013

Benchmark arithmetic and comparison scope

Separate steady-state throughput from startup. Compare only equivalent windows and report the hybrid boundary.

MATCHED FAST BENCHMARK (JOB 37941875)

- Exclusive V100 gpu012, Xeon Gold 6248; Fortran twice, 1 core; then JAX `--rad native`.
- Same nov26 initial state and 108-step window; JAX driver includes per-step state dump.
- Fortran main loop: 246.7 / 246.6 s; mean 2.285 s/step.
- JAX steps 2-107: 55.6 s; mean 0.524 s/step, median 0.456 s.

WHAT THE 4.4x DOES NOT INCLUDE

- Step 0: 933 s; step 1: 96 s. JAX total over 108 steps ~1,085 s vs Fortran ~247 s.
- Steady savings: 2.285 - 0.524 = 1.761 s/step. Estimated crossover ~580 steps (~12 days); assumes constant steady cost and one cold compile.
- Per-step mean includes radiation cadence and boundary work from steps 2-107; median is not the mean.
- No persistent XLA cache (D175).

Replication caveat: one exact matched fast run. The separate 0.434 s mean / profile-wrapper run is corroborating but not a second matched benchmark. Release timing was repeated on two nodes and shows near parity.

Terminology to use

“4.4x steady-state on one optimized V100 configuration” is supported. “4.4x faster model run” is not: startup dominates this 108-step run, and the fast-path long-run workload is untested.

Sources: SPEED_STEP_TIME_LOG_20261012 s13; SPEED_NEXT_STEPS_20261013; RELEASE_NATIVE_RADIATION_20261012

Why it got faster: remove the measured bottlenecks

The fast configuration is a bundle of changes, not one JAX kernel optimization.

Stage / change	What changed	Evidence / caveat
Convection	MSTCNV moved to host Numba worker; Intel libimf math; batched host/device fields.	57/57 fields match NumPy-IMF on captured columns; worker env and startup cost.
Dynamics + ocean	Q-dynamics vmap, pole buffers/async, sequential sum blocks; avoid slower scan unroll.	Targeted checks; flags are opt-in.
Land / Ent	Batch leaf evaluations across qsimp levels and sun/shade cohorts.	Ent call 5.39 -> 2.37 ms; GHY piece 0.0534 -> 0.0296 s; tiny FP-rounding differences.
Host glue / transfers	JIT template-state glue; one batched device_get/put for required MSTCNV outputs.	MSTCNV stage 0.087 -> 0.071 s.
Native radiation	SOCRATES in coupled step; about 0.16-0.17 s per call in isolated V100 test.	Matched native integration released near parity; no longer dominant cost.

Residual fast-step stages (profile run): surface 0.086 s, MSTCNV 0.071 s, dynamics 0.065 s, ocean_b 0.048 s; plus ~0.16 s on radiation steps. These are measured profiles, not a full attribution of every benchmark second.

Sources: SPEED_STEP_TIME_LOG_20261012 s11-13; SOCRATES_R2_DUMPS s12

Fidelity, standalone behavior and test status

Each statement has a scope; do not let a relaxed criterion or successful run imply broader validity.

FIDELITY AGAINST REAL-MEMBER SPREAD

- Fast run: day-1 QCI at step 3 = 2.47x largest member distance; baseline = 2.37x. Strict 2x day criterion: NOT MET.
- Day 2 worst ratio = 1.11 in profile run; relaxed rule (3x for steps >=3) met.
- Owner accepted the ~0.1 ratio shift for the speed experiment; this is not approval of strict fidelity.

LONG-RUN EVIDENCE IS NOT FAST-PATH EVIDENCE

- Release/native non-fast configuration completed a 5-day run without per-step records.
- 30-day shadow run: QDYNAM extra-column branch made states invalid after step 1288 (~26.8 days); 30 days not demonstrated.
- The fast flags have only been exercised for 108 steps (2 days).

TESTS

- Full CPU regression used flags off: 3,329 passed, 1 known order-dependent failure, 16 skipped.
- The flagged configuration has not run the full suite; unit / targeted tests cover selected flags.

BOUNDARY / REPRODUCIBILITY

- nov26 only; five step-0 recorded dumps; irrigation assumptions as documented.
- No other start date, fast-path multi-week run, second exact benchmark, or packaged worker environment yet.

Sources: SPEED_STEP_TIME_LOG_20261012 s13-14; ACCEPTANCE_CRITERIA s9-10; STANDALONE_INPUT_AUDIT s10; SOCRATES_TRIAL_RUN_RECIPE s10

Technical next steps and go / no-go gates

Preserve the release baseline; qualify the fast candidate with evidence matched to the claim.

1	**REPEAT PERFORMANCE**	Same exclusive-node recipe, exact fast flags, same workload. Publish spread, total wall and amortization.
2	**RUN THE FAST TEST MATRIX**	Full regression with flags enabled where applicable; capture environment, worker version and known order-dependent failure.
3	**QUALIFY SCIENCE**	Score against the agreed strict rule; keep any experimental relaxation separate and dated; assess the QCI step-3 shift.
4	**EXTEND THE RUN**	Run fast flags beyond 2 days; debug qdynam_do_z_extra and assess states after that branch before claiming longer free runs.
5	**PROMOTE ONLY IF GATES PASS**	Then decide merge/release; otherwise retain the stable near-parity release and document the fast path as experimental.

Management dependencies: target run length / cold-start tolerance; owner-approved fidelity bar and extra dates; V100 allocation; Numba worker packaging; merge authority.

Sources: SPEED_NEXT_STEPS_20261013; SPEED_RUN_RECIPE_20261013; ACCEPTANCE_CRITERIA